

Course Syllabus

EE 134 Digital Integrated Circuit Design

Winter 2026
University of California, Riverside

Instructor Information

Instructor
Prof. Wantong Li

Email
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Course Description

Lecture: 8:00 am to 9:20 am, Monday and Wednesday

Discussion: 4:00 pm to 4:50pm, Tuesday

Course Overview: Transistor-level analysis and design of digital integrated circuits. Topics include modern transistor devices for digital integrated circuits; in-depth analysis of inverters and logic gates; static and dynamic logic; power analysis of digital integrated circuits; sequential and memory circuits. Uses computer modeling tools in homework assignments to analyze digital integrated circuits.

Prerequisites: EE 100A; EE 120A or CS 120A; or consent of instructor.

Optional Text: *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th Edition, Weste and Harris, Addison-Wesley.

Key Learning Outcomes

1. Analyze functionality of digital ICs including combinational, sequential, and memory types.
2. Characterize speed, power consumption, and robustness of digital ICs.
3. Design digital ICs to meet specified functionality, speed, energy, and robustness targets.
4. Perform simulation of digital ICs and write reports adhering to technical writing standards.

Grading

Midterm Exam	20%
Final Exam	30%
Quizzes	20% (4% each)
Homework 1	10%
Homework 2	10%
Homework 3	10%

Course Schedule

Week	Date	Topic	Homework	Quizzes
1	Jan 05 Jan 07	Course Logistics Overview of Digital IC		
2	Jan 12 Jan 14	MOSFET CMOS Inverter (1) & LTspice		
3	Jan 19 Jan 21	<i>MLK Day</i> CMOS Inverter (2)		Quiz 1 due
4	Jan 26 Jan 28	CMOS Logic Gates Logic Circuits	Homework 1 due	Quiz 2 due
5	Feb 02 Feb 04	Power Analysis Sequential Elements		Quiz 3 due
6	Feb 09 Feb 11	Review Midterm Exam		
7	Feb 16 Feb 18	<i>President's Day</i> Timing Analysis	Homework 2 due	
8	Feb 23 Feb 25	SRAM (1) SRAM (2)		Quiz 4 due
9	Mar 02 Mar 04	DRAM Low-power Design		Quiz 5 due
10	Mar 09 Mar 11	Modern Transistor Scaling Review & Wrap-up	Homework 3 due	
11	TBD	Final Exam		

Exams and quizzes will be administered online via Canvas. Homework must be typed (no handwriting), and will be submitted to Canvas.

Course Expectations and Guidelines

Academic Accommodations: If you are a student with learning needs that require special accommodation, please contact the SDRC office (Student Disability Resource Center, 125 Costo Hall, 951-827-4538, <http://sdrc.ucr.edu/>) as soon as possible, to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail me as soon as possible in order to set up a time to discuss your learning needs.

Academic Rights and Responsibilities: All students, faculty, and staff are responsible for understanding and complying with the University's stated academic requirements. Students should feel free to express their thoughts and opinions in an academic forum. Assignments must be completed by the student for whom the work is assigned and without unauthorized aid of any kind.

Academic Integrity: At UCR we are committed to upholding and promoting the values of the Tartan Soul: Integrity, Accountability, Excellence, and Respect. As a student in this class, it is your responsibility to act in accordance with these values by completing all assignments in the manner described, and by informing the instructor of suspected acts of academic misconduct by your peers. By so doing, you will not only affirm your own integrity, but also the integrity of the intellectual work of this University, and the degree which it represents. Should you choose to commit academic misconduct in this class, you will be held accountable according to the policies set forth by the University, and will incur appropriate consequences both in this class and from Student Conduct and Academic Integrity Programs. For more information regarding University policy and its enforcement, please visit: conduct.ucr.edu.

Classroom Technology Policy: Students are not allowed to use cellphones during lectures, unless in emergencies or directed by the instructor. Laptops and tablets are permitted only for note-taking purposes. During exams, no electronic devices except for non-graphing calculators are permitted.

Collaborations and Group Work: All work in this course require individual work and no collaboration is allowed.

Extensions and Late Assignments: The extension of the deadlines for homework and quizzes, if any, will be announced by the instructor. Assignments late by up to 24 hours will have 20% deducted. Assignments late by up to 48 hours will have 50% deducted. Assignments submitted beyond 48 hours will not be accepted. The exam schedules, however, are strictly enforced.